



Institute of Space Technology

QUALITY SYSTEM PROCEDURE

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THESIS GUIDE

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28 APR 2014

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Vice Chancellor
Institute of Space Technology
(Imran Rahman)

VC Secretariat
Diary No. 562-6
Date: 28/04/14

**AMENDMENT SHEET**

AM D. #	DATE	INITIATED BY	PAGE #	NATURE OF AMENDMENT	DONE BY	APPROVED BY
01	21/04/2014	DQAC	2	Revision of para 5.3	DQAC	

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1 TITLE

THESIS GUIDE

2 PURPOSE

The purpose of this procedure is to provide a system and instructions, and to assign responsibilities for preparation of thesis.

3 SCOPE

This procedure applies to preparation of theses at all levels.

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4 RESPONSIBILITIES

- Students will be responsible to prepare their thesis according to the instructions given in this Thesis Guide.
- Thesis supervisor / HOD of concerned department will be responsible for checking that thesis has been prepared according to the given instructions.
- HOD of concerned department is responsible to keep one copy of thesis with him/her for future record and send one copy of thesis to library for reference.

5. PROCEDURE

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5.1 PAPER

The thesis must be printed on good quality white paper (80 grams) on one side of the paper only. Letter (8.5" x 11") size paper must be used in portrait orientation (vertical orientation).

5.2 MARGINS

Left-hand margin should be 1.5" wide to facilitate binding while right, top and bottom margins should be one inch.

5.3 BINDING

Thesis should be submitted in hard binding according to the following specifications. Text on title page of the thesis should be used for preparing the binding (For sample cover page and spine label, see page one and page two of Thesis Template IST-F-24 respectively). Color of embossed text on binding should be SILVER for BS programs and GOLDEN for MS and PhD programs. Color of binding itself should be according to the following color scheme:

- | | | |
|----|------------------|-----------------------|
| a. | Bachelor Program | Dark Blue (IST Color) |
| b. | Master Program | Maroon |
| c. | PhD Program | Black |

5.4 FONT, HEADINGS, AND LINE SPACING

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For the main body of the text, a standard Times New Roman font, with 12 point font size and double line spacing, should be used throughout the thesis except where instructed otherwise. Headings should stand out clearly from the text. Chapter headings and section headings should be in 14 and 12 point font size respectively. The title of thesis should be in UPPER CASE while remaining headings should be in Title Case and Sentence Case. Samples of heading styles are given in section 5.7.2.1. Footnotes should be 10 point but references or bibliography should be in 12 point font size. The thesis must be printed on a laser printer using black ink. For illustrations, colour printing can be used.

5.5 PAGE NUMBERS

All pages must be numbered in sequence except title page. There must be no missing, blank, or duplicate pages. Specific page number formats are indicated in section seven. Chapter one must start on page 1.

5.6 FOOTNOTES

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Footnotes should be numbered consecutively in text as superscript and detail of footnote should be given at the bottom of the page. Use Times New Roman font with 10 font size. For example:

¹Charles Colson, Being Light In Darkness, (Dallas, TX: Word Publishing, 1992), p. 30.

²Marvin C. Pate, Engineering Design, (Grand Rapids, MI: CRC Press, 1996), p. 95.

5.7 ORDER AND FORMAT OF ITEMS IN THE THESIS

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The following order and format of items should be used in thesis:-

5.7.1 PRELIMINARY PAGES

Preliminary pages should be numbered with lower case Roman numerals using Time New Roman font and 12 point font size at the center aligned bottom of the page.

Example:

ii

5.7.1.1 TITLE PAGE

For sample Title page see page number three of Thesis Template IST-F-24.

5.7.1.2 APPROVAL PAGE

For sample Approval page see page number four of Thesis Template IST-F-24.

5.7.1.3 CERTIFICATE

For sample certificate see page number five of Thesis Template IST-F-24.

5.7.1.4 COPYRIGHT PAGE

For sample copyright page see page number six of Thesis Template IST-F-24.

5.7.1.5 DEDICATION PAGE (OPTIONAL)

You can dedicate your thesis to anyone. A dedication is simply a statement of who the thesis is dedicated to and you can or cannot give a reason.

5.7.1.6 ABSTRACT



Theses must have an abstract. Abstract requirements for theses may vary from department to department. Please check with your advisor if you have any questions about the length of abstract.

5.7.1.7 TABLE OF CONTENTS

For sample Table of Contents see page number seven of Thesis Template IST-F-24.

5.7.1.8 LIST OF TABLES (IF APPLICABLE)

For sample List of Tables see page number eight of Thesis Template IST-F-24.

5.7.1.9 LIST OF FIGURES (IF APPLICABLE)

For sample List of Figures see page number nine of Thesis Template IST-F-24.

5.7.1.10 LIST OF ABBREVIATIONS AND SYMBOLS USED

List can be prepared in single column or two columns but it should be sorted alphabetically by abbreviations. For sample List of Abbreviations and Symbols see page number 10 of Thesis Template IST-F-24.

5.7.1.11 ACKNOWLEDGEMENTS (OPTIONAL)**CONTROLLED**

You can acknowledge financial, moral or physical help provided by your family, friends or anybody else during your thesis / research work but it is not compulsory.

28 APR 2014**5.7.2 MAIN PAGES**

Divide body of the text of your thesis into chapters. Pages should be numbered consecutively in Arabic numerals in Times New Roman font with 12 font size. Page numbers should be given in centre at bottom of the page. For example:

56

5.7.2.1 HEADING STYLE

a) The numbering for chapters should be in Arabic numerals as per example given below. Text of the chapter titles should be bold, centered, upper case, Times New Roman font with 14 points font size.

1. CHAPTER TITLES SHOULD BE BOLD, CENTERED, UPPER CASE, TIMES NEW ROMAN, 14 POINTS

b) The numbering for sections should be in Arabic numeral as per example given below. The first letter in each word of the secondary heading should be capitalized. Secondary heading should be flush left and bold with 12 points font size.

1.1 SECONDARY HEADINGS SHOULD BE FLUSH LEFT, BOLD, 12 POINTS

c) The numbering for third level sections should be in Arabic numerals as per example given below. Only the first letter of the first word of the heading should be capitalized. Third level heading should be flush left and bold with 12 points font size.

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1.1.1 THIRD LEVEL HEADINGS SHOULD BE FLUSH LEFT, BOLD, 12 POINTS

5.7.2.2 TABLES

Tables should be centered between the left and right margin with their captions centered above the table. Tables should be consecutively numbered per chapter. The word Table should be spelled out (not abbreviated) on a line by itself with the table number, and the text of the caption should be centered in point size 12, Times New Roman.

Example:

Table 1.1
The caption should come before the table.

S/No	A	B	C
1	2.2	2.5	5.9
2	1.5	1.6	36

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5.7.2.3 FIGURES

Figures should be centered between the left and right margin with their captions centered below the figure in point size 12 Times New Roman. Figures should be consecutively numbered per chapter. While numbering the figures, the word Figure should be abbreviated as "Fig.".

Example:

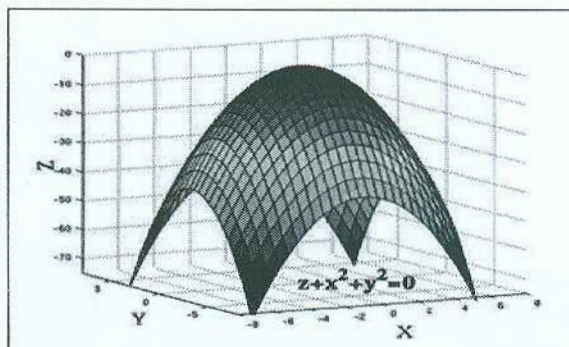


Fig. 1.1. The caption should come under the figure.

5.7.3 FINAL PAGES

Final pages should be numbered in Arabic numerals continuing from the last page number of the previous section of thesis.

5.7.3.1 REFERENCES

References should be numbered consecutively as they appear in the text. Detailed reference should be given on a separate page at the end of the text, under the **References** heading using IEEE Citation Style Guide. Sample references are given below for your convenience:-

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5.7.3.1.1 PRINT REFERENCES

Book

Author(s). Book title. Location: Publishing company, year, pp.

Example:

W.K. Chen. Linear Networks and Systems. Belmont, CA: Wadsworth, 1993, pp. 123-35.

Book Chapters

Author(s). "Chapter title" in Book title, edition, volume. Editors name, Ed. Publishing location: Publishing company, year, pp.

Example:

J.E. Bourne. "Synthetic structure of industrial plastics," in Plastics, 2nd ed., vol. 3. J. Peters, Ed. New York: McGraw-Hill, 1964, pp.15-67.

Article in a Journal

Author(s). "Article title". Journal title, vol., pp, date.

Example:

G. Pevere. "Infrared Nation." The International Journal of Infrared Design, vol. 33, pp. 56-99, Jan. 1979.

Articles from Conference Proceedings (published)

Author(s). "Article title." Conference proceedings, year, pp.

Example:

D.B. Payne and H.G. Gunhold. "Digital sundials and broadband technology," in Proc. IOOC-ECOC, 1986, pp. 557-998.

Papers Presented at Conferences (unpublished)

Author(s). "Paper's title," Conference name, Location, year.

Example:

B. Brandli and M. Dick. "Engineering names and concepts," presented at the 2nd Int. Conf. Engineering Education, Frankfurt, Germany, 1999.

Standards/Patents

Author(s)/Inventor(s). "Name/Title." Country where patent is registered. Patent number, date.

Example:

E.E. Rebecca. "Alternating current fed power supply." U.S. Patent 7 897 777, Nov. 3, 1987.

5.7.3.1.2 ELECTRONIC REFERENCES

Book

Author. (year, Month day). Book title. (edition). [Type of medium]. Vol. (issue).

Available: site/path/file [date accessed].

Example:

S. Calmer. (1999, June 1). Engineering and Art. (2nd edition). [On-line]. 27(3). Available: www.enggart.com/examples/students.html [May 21, 2003].

Journal



Author. (year, month). "Article title." Journal title. [Type of medium]. Vol. (issue), pages.
Available: site/path/file [date accessed].

Example:

A. Paul. (1987, Oct.). "Electrical properties of flying machines." Flying Machines. [Online]. 38(1), pp. 778-998. Available: www.flyingmachjourn/properties/fly.edu [Dec. 1, 2003].

World Wide Web

Author(s)*. "Title." Internet: complete URL, date updated* [date accessed].

Example:

M. Duncan. "Engineering Concepts on Ice. Internet: www.iceengg.edu/staff.html, Oct. 25, 2000 [Nov. 29, 2003].

5.7.3.1.3 ODD SOURCES

CONTROLLED

Newspaper

Author(s)*. "Article title." Newspaper (month, year), section, pages.

Examples:

B. Bart. "Going Faster." Globe and Mail (Oct. 14, 2002), sec. A p.1.
"Telehealth in Alberta." Toronto Star (Nov. 12, 2003), sec. G pp. 1-3.

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Dissertations and Theses

Author. "Title." Degree level, school, location, year.

Example:

S. Mack. "Desperate Optimism." M.A. thesis, University of Calgary, Canada, 2000.

Lecture

Lecturer(s). Occasion, Topic: "Lecture title." Location, date.

Example:

S. Maw. Engg 251. Class Lecture, Topic: "Speed skating." ICT 224, Faculty of Engineering, University of Calgary, Calgary, Alberta, Oct. 31, 2003.

E-mail

Author. Subject line of posting. Personal E-mail (date).

Example:

J. Aston. "RE: new location, okay?" Personal e-mail (Jul. 3, 2003).

Internet - Newsgroup

Author or Topic*, "Title," Complete network address, date when it was updated [date accessed].

Example:

G.G. Gavin. "Climbing and limb torsion #3387," USENET: sci.climb.torsion, Apr. 19, 2000 [Oct. 4, 2002].

* if you can't find this information, exclude it.

Exact page number References

To refer readers to specific page numbers in a text, use the number of the reference followed by a colon (:) and the page numbers.

Example:

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Johnson suggests that citing will lead to a decrease in being cited for plagiarism [1:28-29].

The [1] refers to the numbered reference and the 28-29 refers to the pages being cited.

5.7.3.2 BIBLIOGRAPHY (OPTIONAL)

If you want to provide bibliography of related literature, in addition to the literature referred to in the body of thesis text, you can do so but it is not compulsory.

5.7.3.3 APPENDICES (OPTIONAL)

If you want to provide additional / related material, which may be helpful in understanding the text of your thesis, you can do so in the form of appendices but these are not compulsory. Appendices should be numbered with English Alphabets on right side of the page.

Example:

CONTROLLED

Appendix – A

5.8 ELECTRONIC SUPPLEMENTS

Copy of thesis (full text), with all relevant materials, should be provided on a CD/DVD along with the thesis. CD/DVD should be labeled with student's name, thesis title, institution, date etc., and should be placed in a pocket attached to the inner side of back cover.

5.9 CHECKLIST

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Please check your thesis in the light of following checklist before submission. This checklist will help you in deciding whether your thesis is ready for submission or not. If answer to any one of following questions is in NO then your thesis is not yet ready for submission.

- Have you prepared your thesis according to the instructions given in this guide?
- Have you formatted your thesis according to the instructions given in this guide?
- Have you numbered all pages of your thesis according to the instructions given in this guide?
- Have you placed all items in your thesis in correct order according to prescribed format?
- Have you printed your thesis on good quality white paper (80 grams)?
- Have you got your thesis bound according to the sample provided in this guide?
- Have you chosen right colours for binding according to your department?
- Have you obtained signature on the certificate page from your supervisor?
- Have you attached an electronic copy of thesis and related files in every copy of your thesis?
- Was your thesis processed for plagiarism checking?

6 RELATED FORMS AND TEMPLATES

Form Number	Form Name	Form Location (Path)	User
IST-F-24	Thesis Template	Intranet	Students

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7 RELATED RECORDS

Record Name	Storage Mode (Soft/Hard)	Responsible	Retention Time
Thesis	Soft / Hard	Library	Forever
Thesis	Soft / Hard	HOD of concerned Department	Forever

CONTROLLED

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CONTROLLER OF SPACE LAUNCH VEHICLE



by
Sunie Hassan
Shaharyar Malik

CONTROLLED
26 SEP 2014

Supervisor
Dr. Sarwar Naqvi

Department of Aeronautics and Astronautics
Institute of Space Technology, Islamabad
2012

One Inch
Blank Space

2012

CONTROLLER OF SPACE LAUNCH VEHICLE

CONTROLLED
26 SEP 2014

Three Inches
Blank Space

CONTROLLER OF SPACE LAUNCH VEHICLE

A thesis submitted to the
Institute of Space Technology
in partial fulfillment of the requirements
for the degree of Bachelor of Science in
Aerospace Engineering

by

Sunir Hassan
Shaharyar Malik

Supervisor
Dr. Sarwar Naqvi

CONTROLLED

26 SEP 2014

Department of Aeronautics and Astronautics
Institute of Space Technology, Islamabad
August 2012

Institute of Space Technology

Department of Aeronautics & Astronautics



Controller of Space Launch Vehicle

by

Sunie Hassan
Shaharyar Malik

APPROVAL BY BOARD OF EXAMINERS

Supervisor's Name

CONTROLLED

26 SEP 2014

Advisor 1 (Name)

Advisor 2 (Name)

AUTHOR'S DECLARATION

I/We take full responsibility of the research work conducted during the FYP, MS/PhD Thesis, published Journal / Conference Paper or any Patent generated titled “ _____ ”.

I solemnly declare that the research and development work presented in the FYP, MS/PhD Thesis, Published Journal / Conference Paper or any Patent generated is done solely by me with no significant help from any other person; however, small help wherever taken is duly acknowledged. I have also written the complete thesis / paper by myself. Moreover, I have not presented this thesis / paper (or substantially similar research and development work) or any part of the thesis / paper previously to any other degree awarding institution within Pakistan or abroad.

CONTROLLED

I understand that the management of IST has a zero tolerance policy towards plagiarism. Therefore, I as an author of the above-mentioned thesis / paper, solemnly declare that no portion of my thesis / paper has been plagiarized and any material used in the Thesis / paper from other sources is properly referenced. Moreover, the thesis / paper do not contain any literal citing (verbatim) of more than 70 words (total) even by giving a reference unless I have obtained the written permission of the publisher to do so. Furthermore, the work presented in the thesis / paper is my own original work and I have positively cited the related work of the other researchers by clearly differentiating my work from their relevant work.

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I further understand that if I am found guilty of any form of plagiarism in my thesis / paper work even after my graduation, the Institute reserves the right to revoke my BS / MS/ PhD degree. Moreover, the Institute will also have

the right to publish my name on its website that keeps a record of the students who plagiarized in their theses /paper work.

Signature

Student Name

Batch-Roll #

I hereby acknowledge that submitted thesis / report/ paper is final version and should be scrutinized for plagiarism as per IST policy.

Supervisor

Dated: _____

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26 SEP 2014

Verified by Plagiarism Cell Officer

Dated: _____

In case, of a multi authored research project the plagiarism undertaking "I" will replaced with "We" in the plagiarism Undertaking.

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SAMPLE

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List of Abbreviations and Symbols

mg/l	Micro gram per liter
ppb	Parts per billion
ppm	Parts per million
SORAS	Solar Oxidation
UN	United Nations

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